Understanding College Enrollment through Causality

Semester 1 Overview

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Section 1

Introduction

Summer 2024

This summer, I developed first-party data utilization initiatives for Dotdash Meredith (People, Entertainment Weekly, SimplyRecipes).

product managers at the technical all hands

Students as Online Advertisements

While at DDM, I discovered Latent Profile Analysis, a statistical method being used to target online ads and segment users by

- Session time
- Page Interactions
- Other invasive metrics...

Underlying belief: There are distinguishable types of users online, which all have different viewing/spending habits and behaviors.

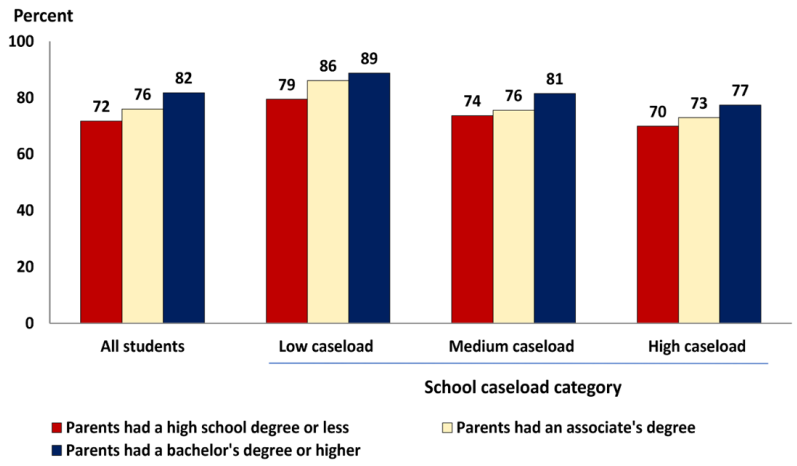
Question: Could LPA be used to segment types of *students* applying to college? If so, would we find similar types of students at the same schools?

College Counseling

College counseling is a focal inequity—half of of all U.S. high school students seek private counseling for college application support.

- Recommended counselor-student ratio, 1:250
- Public School counselor-student ratio, 1:350

Parental Educational Attainment (PEA) and socioeconomic status further exacerbate this reality for disadvantaged students:



National Center for Education Statistics. *How Often Do High School Students Meet with Counselors About College? Differences by Parental Education and Counselor Caseload.* 2024.

A Solution

Counselor contact is a significant indicator of college enrollment. Counselors are best at helping students determine what schools they are *best fit* for.

- Academic Merit: GPA, Test Scores, class Rank, curricular Rigor
- Character: Motivations, personality, interests
- Preference: Campus Size, location, mascot . . .

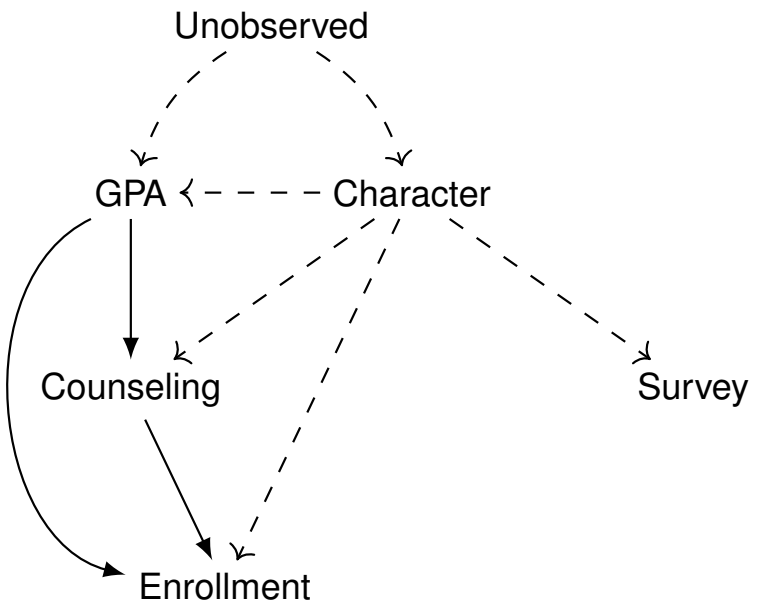
Motivating Question: Could statistical methods like LPA be used to help students understand their fit-factor *holisitcally*?

Overarching Question: How could we estimate the effect of counseling on enrollment? HOw then would we account for character?

College Admissions: A Causal Representation

We believe that factors like GPA, character, and counseling significantly influence one another—ultimately affecting students' enrollment outcomes.

We can represent the *underlying causal nature* of the current college admissions scheme with a causal graph:



Causal Inference

Causal Inference is a framework for estimating the causal effect of "treatment" or "intervention" on "outcome".

Propensity Score Matching enables us to emulate the environment of a randomized controlled trial and reduce bias for observational studies. In our context:

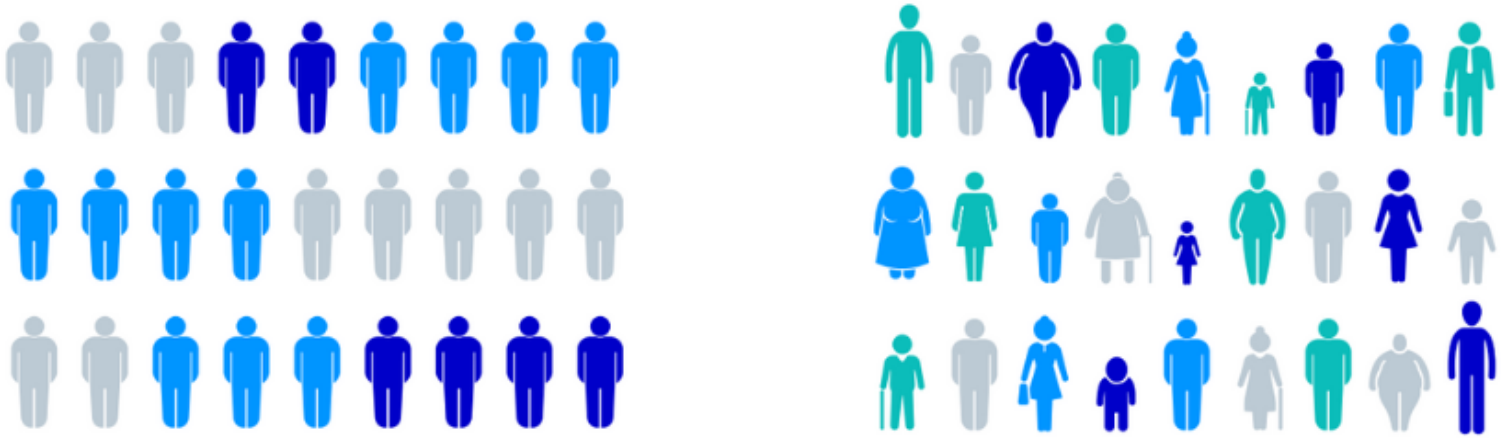
Propensity score
=
probability of student recieveing counseling given their GPA and character

ADJUSTING FOR POTENTIAL CONFOUNDING EFFECTS IN REAL-WORLD STUDIES:

Propensity Score Matching

Pfizer Medical Affairs

CONFOUNDING EFFECTS IN REAL-WORLD STUDIES



Randomized controlled trials

Real-world studies

Motivation

GPA and counseling can be considered *observed* variables, whereas a students character is inherently *unobserved*.

We'd like to understand the relationships of *each* of the variables in the DAG, beginning with counseling and enrollment.

To start, we must find a way to estimate character.

Character: Next Steps

Currently, research on character resides squarely in the domain of psychology. The American Psychology Association's definition of "character" is as follows:

> *The totality of an individual's attributes and personality traits, particularly their characteristic moral, social, and religious attitudes.*

In the next section, we will approach estimating character via

- The Five Factor Model (FFM)
- Narrative Identity

Section 2

Quantifying Character

The Five-Factor Model

The five-factor model (FFM) is a paradigm through which psychologists can classify one's personality using 5 basic traits:

- Neuroticism
- Extraversion
- Openness (to experience)
- Agreeableness
- Conscientiousness

Commonly assessed using surveys that present participants with a series of prompts designed to measure specific traits.

Ex: ***On a scale from 1-5, how strongly do you agree with the following statement:*** *I find the world a very interesting place.*

Major Flaw: Given time and sufficient motivation, people can change!

Narrative Identity

Narrative identity: An individual's reconstructed past, perceived present, and imagined future, internalized as the the story they use to define themselves both to others and themselves.

Ex. **Describe your relationship to learning and education:**

Since I was a kid, I always had people to work with on things that challenged me, but also on things that excited me. As I grew up, grades became more important to me, but looking back on it, I suppose being open to collaboration helped me to thrive academically.

Major Flaw: Difficult to quantify, and even harder to scale.

Structural Topic Modeling

Topic modeling in natural language processing is a text mining technique that aims to identify keywords and common phrases in large text data sets, or corpora.

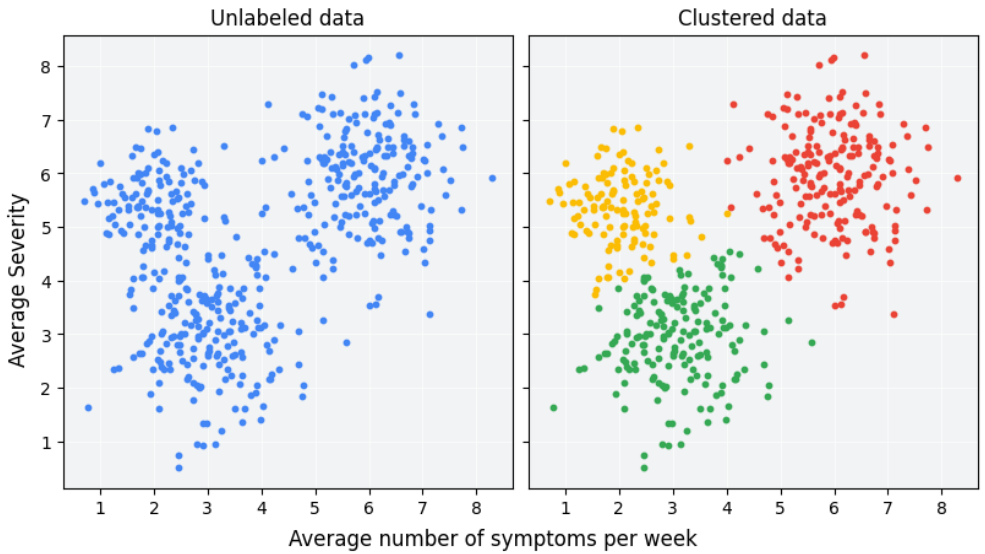
Terms	Documents								
	c1	c2	c3	c4	c5	m1	m2	m3	m4
human	1	0	0	1	0	0	0	0	0
interface	1	0	1	0	0	0	0	0	0
computer	1	1	0	0	0	0	0	0	0
user	0	1	1	0	1	0	0	0	0
system	0	1	1	2	0	0	0	0	0
response	0	1	0	0	1	0	0	0	0
time	0	1	0	0	1	0	0	0	0
EPS	0	0	1	1	0	0	0	0	0
survey	0	1	0	0	0	0	0	0	1
trees	0	0	0	0	0	1	1	1	0
graph	0	0	0	0	0	0	1	1	1
minors	0	0	0	0	0	0	0	1	1

Structural topic modeling is a modification which enables us to incorporate document meta-data (respondent characteristics, time).

Latent Profile Analysis

Latent profile analysis is a clustering method used to reveal unobserved groups—or latent profiles—in data, among a set of observed, continuous variables. Latent *class* analysis is the categorical analog.

Think: Types of patients upon taking Tylenol.



Google Developers. Clustering: Overview. Accessed 2024.

Approach 1: FFM

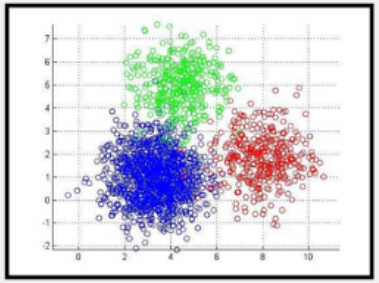
Data Aggregation

Survey ***n*** students by multiple choice, with ***p*** prompts, each assessing a different trait.

Survey Result Wrangling

Treat each prompt like a variable, and every respondent as an observation.

This way, we can work with the survey like we would an observational setting.



Response Clustering

With LCA, we group respondents by their response patterns.

→

Students with similarly expressed traits.

Approach 2: Narrative Identity

Data Aggregation

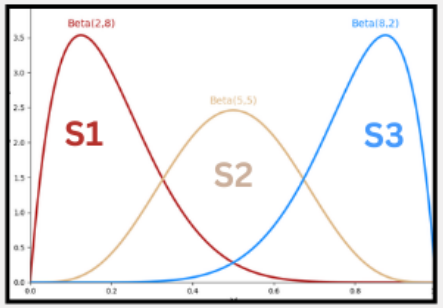
Survey ***n*** students by short response, with ***p*** prompts, each assessing a different trait.

	Topic 1	Topic 2	Topic 3
Doc 1	.95	.08	.64
Doc 2	.58	.42	.01
Doc 3	.02	.01	.36

Topic Association

STM processes text responses and creates a table of probabilities.

We will assume that *each row* of probabilities is derived from a ***beta distribution.***



Distributional Clustering

With LPA, we group documents by their probability distributions across topics.

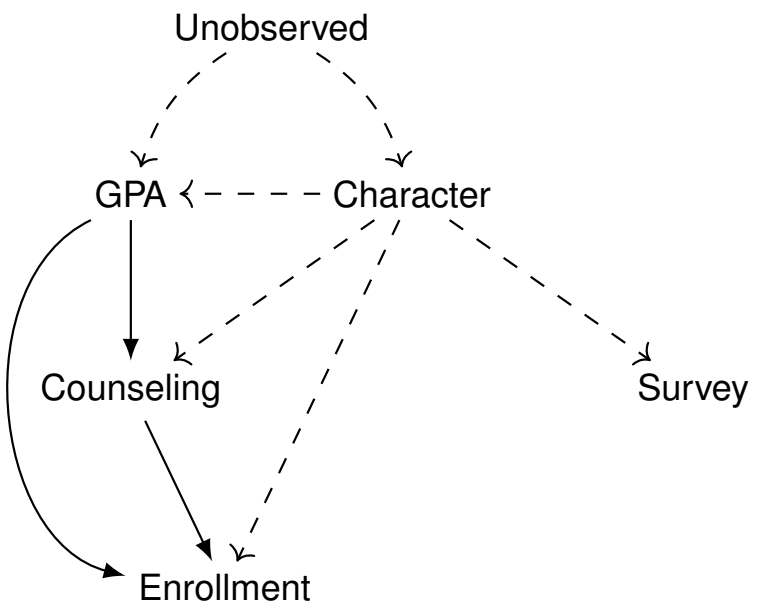
→

Students with similarly expressed traits.

Section 3

Simulation

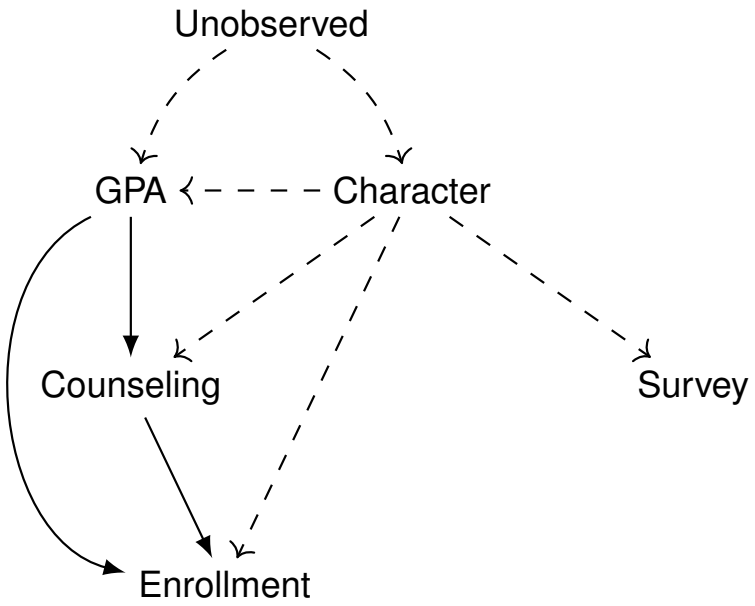
Data-Generating Process



We define the variables in our DAG as such:

- $U \sim \mathcal{N}(0, 1)$, `True_Char` = $\{1, 2, 3, 4\}$
- $GPA \sim \mathcal{N}$, ranging from 2.3 to 3.9 roughly—a function of U and `True_Char`.

Data-Generating Process (cont.)



- $C = \{0, 1\}$, binary status indicator—a function of *GPA* and `True_Char`.
- $E = \{0, 1\}$, binary status indicator—**only** a function of *GPA* and `True_Char`.

The purpose of this simulation is to test the efficacy of our character estimation methods.

FFM Approach: `Est_Char`

Our character estimates will be the outputs of our LCA algorithm under two different settings:

- Easy: The chance of a student picking a response perfectly in-line with their true character is very high.
- Hard: Students responses are less consistent with their true character.

Ex:

- Easy: If a student is of true character (1), then they are 70% more likely to choose response 1 for each question than any other response.

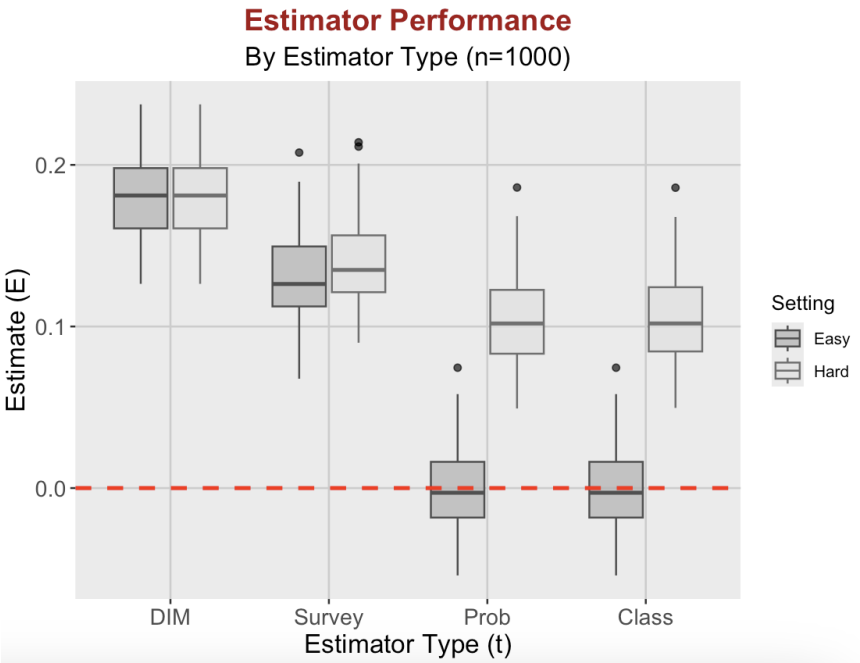
- Hard: If a student is of true character (1), then they are 20% less likely to choose response 1 for each question than any other response.

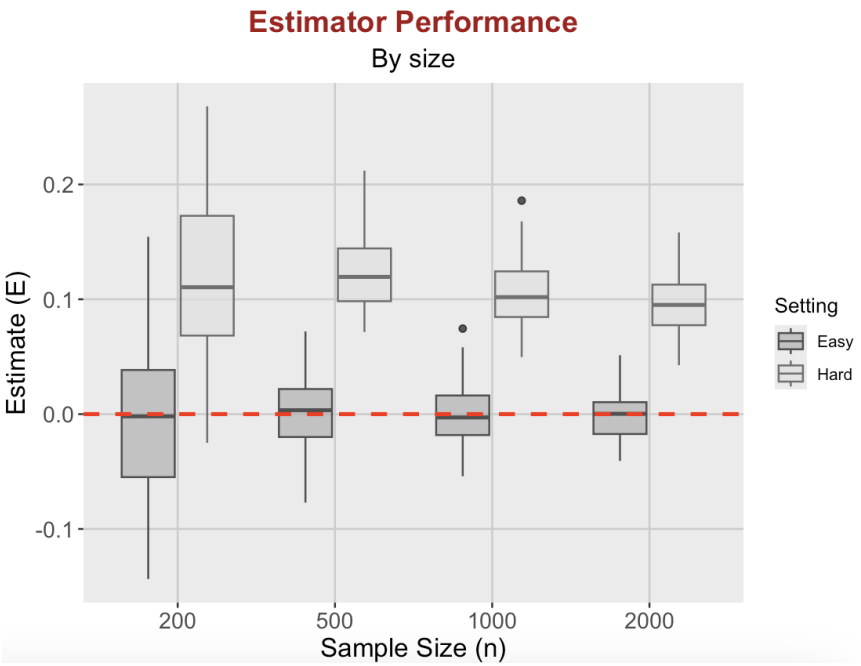
Estimators (pt.1)

The following are what estimators we will use to glean the causal effect (0) of counseling on enrollment:

- **Difference-in-Means (DIM)**: The difference between the average enrollment in the treated group and control groups (with or without counseling).
- **PS Classes**: Using LCA determined classes to match control and treatment groups.
- **PS Probabilities**: Using LCA determined posterior probabilities to match control and treatment groups.
- **PS Survey Results**: Using the bare survey results for matching.

Results (pt.1)





- LCA results results gave us better estimators than survey or naive DIM estimators.
- Easy setting estimators are more unbiased on average, hard estimators struggle even at larger sample sizes.

Narrative Identity Approach: `Est_Char`

Our character estimates will be the outputs of our LPA algorithm under two different settings:

- Easy: Respondent's document-topic probability distributions will have little overlap.
- Hard: Distributions will be almost identical.

Ex:

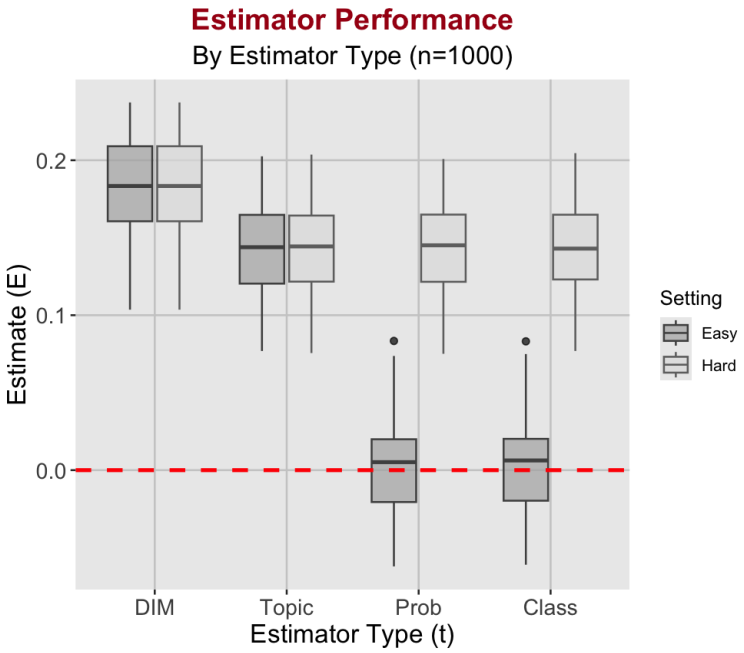
In a hard setting, students will write responses in similar ways, and about similar things.

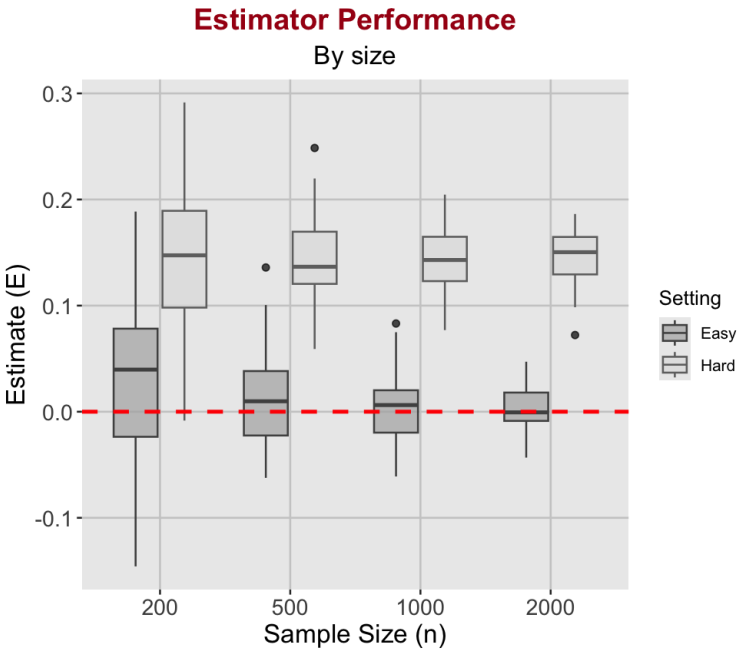
Estimators (pt.2)

The following are what estimators we will use to glean the causal effect (0) of counseling on enrollment:

- **Difference-in-Means (DIM)**
- **PS Classes**
- **PS Probabilities**
- **PS Top Topics**: Using each document's top topic (by probability) for matching.

Results (pt.2)





- LPA results results gave us better estimators than topic or naive DIM estimators.
- Easy and hard setting estimators nearly identical when using top topics.

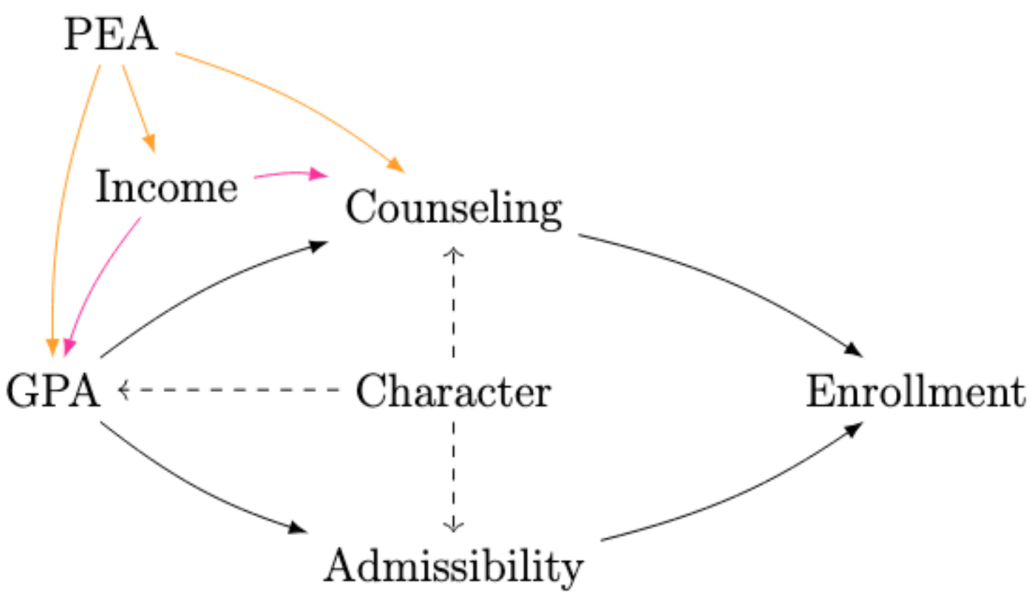
Section 4

Next Steps

But wait. . .

How could

- Admissibility be used as a proxy for enrollment?
- Socio-economic and parental education status be incorporated as confounders?
- Student longitudinal data be involved in character estimation?



Thank You

Questions?